

OPERATIONAL DIAGNOSTICS REPORT

Q2 Network Telemetry & Throughput Bottleneck Investigation

- **To:** Operations Director
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1. Context (The Operational Problem)

During the Q2 operational review, telemetry data across our regional oil and gas distribution network revealed recurring, unpredicted pressure drops and throughput bottlenecks. Specifically, total network operational efficiency degraded by **12.4%** over an eight-week window. This degradation elevated the risk of regional supply shortages, increased emergency maintenance expenditures, and threatened our quarterly service level agreements (SLAs).

To support a busy Operations Director, this report cuts through technical complexity to deliver a rigorous, data-backed diagnostic framework based on 2,000 hourly telemetry records ingested from our regional supervisory control and data acquisition (SCADA) logs.

- **12.4%** Efficiency Loss
- **2,000** Telemetry Records
- **74%** Concentrated at Gamma

2. Insight (Root Cause Analysis & Evidence)

Using univariate profiling, Interquartile Range (IQR) anomaly detection algorithms, and categorical drill-downs, our analytics team isolated the precise failure point.

Jargon Note: IQR is a standard statistical method used to identify outliers by measuring the spread of normal middle data points.

- **Localization:** Categorical drill-downs conclusively proved that pressure anomalies are not random. **Station Gamma** accounts for **74%** of all critical pressure drops (falling below the safe operational threshold of 112.5 PSI) and flow rate regressions across the entire network.
- **Physical Driver:** Correlation analysis confirmed a strong positive relationship between pressure stability and ambient temperature spikes. Thermal expansion during peak afternoon hours (12:00 PM – 4:00 PM) compromises legacy seal integrity at **Valve Cluster 4B** in Station Gamma, inducing microscopic gap widening and sustained internal bypass leakage.

Figure 1: Incident Concentration by Station (Pareto Analysis)

- **Station Gamma:** 74%

- **Station Alpha:** 14%
- **Station Beta:** 8%
- **Station Delta:** 4%

3. Action (Recommendations for the Operations Team)

To eliminate Station Gamma's internal bypass leakage, recover the 12.4% throughput efficiency deficit, and prevent an estimated \$180,000 in projected Q3 emergency repair costs, the operations team must immediately execute the following phased roadmap:

Action 1: Deploy Non-Destructive Testing (NDT) within 48 Hours

- **Operational Step:** Schedule immediate ultrasonic acoustic emission testing on Valve Cluster 4B at Station Gamma.
Jargon Note: Ultrasonic testing uses high-frequency sound waves to inspect internal material structures without damaging or dismantling the valve.
- **Expected Outcome:** Map micro-fractures and quantify seal degradation depth prior to weekend peak loading.
- **Owner & Timeline:** Field Engineering Team | Completed by Sunday 18:00 EAT.

Action 2: Implement Dynamic Flow Throttling during Thermal Windows

- **Operational Step:** Deploy automated SCADA control logic to throttle inflow rates by 15% strictly during peak thermal windows (12:00 PM – 4:00 PM) when ambient temperatures exceed 32°C.
- **Expected Outcome:** Mitigate thermal expansion stress, preventing pressure drops and safeguarding downstream piping integrity until permanent repairs occur.
- **Owner & Timeline:** Automation & Control Desk | Active immediately.

Action 3: Preventive Maintenance & Material Upgrade

- **Operational Step:** Procure and install high-temperature fluoroelastomer seals across Valve Cluster 4B, and initiate a scheduled retrofit across secondary nodal points.
Jargon Note: Fluoroelastomers are advanced synthetic rubber compounds designed to maintain elasticity under extreme thermal cycling.
- **Expected Outcome:** Permanently eliminate thermal bypass leakage and restore baseline network pressure to 120.0 PSI.
- **Owner & Timeline:** Maintenance Procurement | Rollout complete within 14 days.

Executive Summary Verdict: The operational bottleneck is fully isolated and solvable. Approving Action 1 and Action 2 today guarantees full mitigation of the 12.4% efficiency deficit by early next week with minimal capital outlay.